

AWS Solutions Architect

Associate · SAA-C03 · Recall Poster

How to use this

Left of each row is the **cue**, right is the **answer**. Lay a ruler down the middle of a column, work down saying answers out loud, then slide it away to check. Blocks are grouped by area, so you can drill one area at a time. Amber strips are **traps** — where a plausible-looking answer is wrong.

Marks: asked a lot asked often asked thinly — counted across 522 real practice questions, not guessed. a trap sits here. commonly reworded — see **Also asked as**.

30% Secure	26% Resilient	24% High-performing	20% Cost-optimized
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THE NUMBERS		pure rote — learn these first
S3 max object / single PUT	5 TB / 5 GB	
■■■ Min duration IA / Glacier / Deep +	30 / 90 / 180 days	
S3 multipart part size	5 MB – 5 GB, max 10,000 parts	
■■■ Glacier expedited / standard / bulk	1–5 min / 3–5 h / 5–12 h	
■■■ Lambda max duration +	15 min (900 s)	
■■■ Lambda memory	128 MB – 10,240 MB	
■■■ Lambda payload sync / async	6 MB / 256 KB	
■■■ Lambda default concurrency	1,000 per region	
■■■ SQS max message	256 KB	
■■■ SQS retention default / max	4 days / 14 days	
■■■ SQS visibility default / max	30 s / 12 h	
■■■ SQS long poll wait	20 seconds	
■■ Kinesis shard in / out	1 MB/s / 2 MB/s	
■■ Kinesis retention default / max	24 h / 365 days	
■■■ DynamoDB max item	400 KB	
■■■ DynamoDB RCU / WCU	4 KB strong read / 1 KB write	
RDS backup retention	7 days default, 0 – 35 range	
■■■ DynamoDB LSI / GSI per table	5 / 20	
■■■ Read replicas RDS / Aurora	5 / 15	
■■■ Max storage RDS / Aurora	64 TiB / 128 TiB	
VPC CIDR range	/16 down to /28	
Reserved IPs / usable in a /24	5 / 251	
■■ Spot notice; spot & RI discount	2 min; ~90% / ~72%	
■■ CloudWatch basic / detailed	5 min / 1 min	

EC2		compute
Instance states	pending, running, stopping, stopped, shutting-down, terminated, rebooting	
Survives reboot but not stop	instance store data + public IPv4	
Stop: lost / kept	RAM & instance store / EBS volumes	
Metadata endpoint, version to use	169.254.169.254, IMDSv2	
Hibernate saves	RAM, to the root EBS volume	
User data runs	once, at first boot, as root	
System check vs instance check	AWS problem, stop/start to move host — vs your OS or config	
■■■ Low latency / isolation / distributed	cluster / spread / partition placement group	
■■■ Socket or core-based licence †	Dedicated Host	
■■■ Steady 24/7 vs interruptible batch †≈	Savings Plan or RI vs Spot	
SSH user: Amazon Linux / Ubuntu	ec2-user / ubuntu (CentOS: centos, Debian: admin)	
AMI, ECR image, snapshot scope	region — must be copied elsewhere	
Trap: Dedicated Instance only isolates hardware per account. Only a Dedicated Host gives you the physical server and its socket and core counts.		

AUTO SCALING & LOAD BALANCING	
Default cooldown and grace period	300 seconds each
■■■ Recommended scaling policy	target tracking
Known spike / learned pattern	scheduled / predictive scaling
Instances killed before app boots	raise the grace period; use an ELB health check
Group flapping in and out	raise cooldown, widen the alarm threshold
Script before termination / debug one	lifecycle hook / standby
■■■ Define the instances	launch template (configurations are legacy)
■■■ ALB / NLB / GWLB layer	7 / 4 / 3
■■■ Static IP, UDP, millions of req/s	NLB
■■■ Path or host routing, dynamic container ports, Cognito auth	ALB
■■■ Cross-zone default +	ALB on, NLB off, Classic off
Min subnets, internet-facing	2 subnets in 2 AZs
Health check lives on / many certs	the target group / SNI

LAMBDA, CONTAINERS & APP PLATFORMS	
■■■ Needs longer than 15 minutes ≈	Fargate, ECS, Batch or Step Functions
Faster with no code change ≈	more memory (CPU scales with it)
Cold starts / starving the account	provisioned / reserved concurrency
429 TooManyRequests	concurrency limit hit
■■■ Share dependencies / failed async	layers / dead-letter queue
■■■ Lambda in a VPC needs internet	NAT gateway
■■■ ECS task role vs execution role	your code's permissions vs ECS pulling the image
■■■ Kubernetes or kubectl	EKS
■■■ Containers: no servers / needs GPU	Fargate / EC2 launch type
■■■ Upload code, AWS builds the stack	Elastic Beanstalk (Immutable or Blue/Green = zero downtime)
Chef or Puppet / long batch jobs	OpsWorks / AWS Batch
■■■ Edge: light rewrite / calls services	CloudFront Functions / Lambda@Edge

S3		object storage
Consistency model	strong read-after-write	
■■■ Default encryption today †	SSE-S3, applied automatically	
■■■ SSE-S3 / SSE-KMS / SSE-C	AWS holds all keys / you own the CMK / you send the key every request	
■■■ Accidental vs permanent delete	versioning / MFA Delete (root + CLI)	
■■■ Undeletable for seven years	Object Lock, compliance mode	
■■■ Replication requires †	versioning on both + an IAM role — and it never copies existing objects	
■■■ Slow uploads from far away	Transfer Acceleration	
■■■ Browser uploads one file securely	presigned URL	
■■■ CloudFront reads a private bucket	Origin Access Control (OAC)	
Stop all public access	S3 Block Public Access	
■■■ Hot / unknown pattern / infrequent ≈	Standard / Intelligent-Tiering / Standard-IA	
■■■ Events can go to	SQS, SNS, Lambda, EventBridge	
■■■ Infrequent and re-creatable †	One Zone-IA (single AZ)	
■■■ Archive: ms / minutes–hours / cheapest	Glacier Instant / Flexible / Deep Archive	
Trap: One Zone-IA lives in one AZ. If the question says "must survive the loss of an Availability Zone," it is wrong, however cheap it looks.		

EBS, EFS, FSX & TRANSFER	
EBS scope †	one AZ — never spans AZs
Move a volume to another AZ	snapshot, then create from it there
Encrypt an existing volume or RDS	snapshot → copy with encryption → restore
Changeable while running	type, size, IOPS (grow only, never shrink)
Snapshots stored in	S3, incremental, region-scoped
■ ■ General default volume	gp3 (3,000 IOPS / 125 MB/s included)
■ ■ Guaranteed IOPS / throughput / cold	io2 / st1 / sc1 (st1 & sc1 cannot boot)
Highest IOPS, data disposable	instance store
■ ■ Shared files, Linux, many instances ≈	EFS (NFS v4.1, regional)
■ ■ SMB, Windows, Active Directory ≈	FSx for Windows File Server
■ ■ HPC or ML scratch, links to S3	FSx for Lustre
■ ■ On-prem NFS/SMB backed by S3	S3 File Gateway
■ ■ Shrink on-prem footprint vs keep it all local	Volume Gateway cached / stored mode
■ ■ Retire a tape library	Tape Gateway
■ ■ Petabytes, no bandwidth vs ongoing sync ≈	Snowball / DataSync
One backup policy across services	AWS Backup
DATABASES	

■■■ Multi-AZ purpose †≈	availability — synchronous, not readable
■■ Read replica purpose †≈	read scaling — async, readable, cross-region
■■■ Multi-AZ failover	automatic, 60–120 s, same endpoint
RDS gives you OS access †	no — need it? run on EC2
■■■ Aurora copies / failover	6 across 3 AZs / under 30 s
Restoring a snapshot gives	a new instance with a new endpoint
■■■ Cross-region DR, sub-second lag	Aurora Global Database
■■■ Intermittent load, near-zero idle cost	Aurora Serverless v2
■■■ DynamoDB microsecond reads ≈	DAX (read-through only)
LSI vs GSI	same PK, made with the table / new PK any time, eventually consistent
■■■ Active-active across regions	Global Tables
■■■ Every item change / expire free	DynamoDB Streams + Lambda / TTL
■■ ProvisionedThroughputExceeded	hot partition → better key, or on-demand
■■ Redis vs Memcached	persistence, failover, sorted sets / simple multi-threaded cache
■■ Warehouse vs ad-hoc SQL on S3	Redshift / Athena

Short queries stuck behind long	Redshift WLM query queues
Graph / time series / ledger	Neptune / Timestream / QLDB
Migrate engine, low downtime	DMS + Schema Conversion Tool
Trap: you can never read from a Multi-AZ standby. If an answer offers the standby for reporting queries it is wrong — that is what a read replica is for.	

VPC & CONNECTIVITY	
What makes a subnet public	a route to an internet gateway
Private instances reach the internet ≈	NAT gateway in a public subnet (one per AZ)
Security group	instance level, stateful, allow-only
Network ACL +	subnet level, stateless, allow + deny, in rule order
Block a single malicious IP +	NACL deny rule (groups cannot deny)
Default vs custom NACL +	allows everything / denies everything
Ephemeral port range +	1024 – 65535
Two VPCs vs hundreds +	peering (not transitive) / Transit Gateway
Expose your service to another VPC	PrivateLink + NLB
Gateway endpoints work with +	S3 and DynamoDB only (free)
Interface endpoint	everything else — an ENI with a security group
Hybrid in days vs steady bandwidth ≈	Site-to-Site VPN / Direct Connect (weeks)
Why traffic is being dropped	VPC Flow Logs — look for REJECT
Shell access, no open ports, audited ≈	SSM Session Manager
Trap: NACLs are stateless. An inbound rule alone is never enough — you must also allow the outbound ephemeral ports so the reply can get out.	

ROUTE 53 & THE EDGE	
A/B test, gradual migration	Weighted
Active-passive disaster recovery	Failover
Compliance, language, licensing ≈	Geolocation (the user's location)
Shift load between regions with a bias	Geoproximity (your resources)
Fastest region for each user ≈	Latency
Up to 8 healthy records	Multivalue answer
Works at the zone apex, free +	Alias (a CNAME cannot)
CloudFront default / max TTL	24 hours / 1 year
Health check failure threshold	3 consecutive checks
CloudFront certificate must live in +	us-east-1
Restrict content to paying users	signed URLs or signed cookies
Static anycast IPs, TCP/UDP, gaming	Global Accelerator (does not cache)

SECURITY & IDENTITY		30% of the exam
Policy evaluation order ≈	explicit Deny → explicit Allow → implicit deny	
Never grant, only limit	SCPs, permission boundaries, session policies	
SCP effect on the management account +	none — it is never restricted	
App on EC2 needs S3 ≈	IAM role via an instance profile	
Another account needs access	cross-account role + sts:AssumeRole	
Workforce sign-in / mobile users	IAM Identity Center / Cognito identity pools	
Resource-based policy must have	a Principal	
KMS direct encryption limit	4 KB → above that, envelope encryption	
Single-tenant, FIPS 140-2 Level 3	CloudHSM	
Rotate DB creds / free config store	Secrets Manager / SSM Parameter Store	
SQL injection, XSS, rate limit by IP	AWS WAF	
DDoS Response Team, cost protection	Shield Advanced	
GuardDuty / Inspector / Macie	detects threats / scans for CVEs / classifies PII	
Detective / Security Hub	investigates / collects all findings	
Trusted Advisor categories	Cost, Performance, Fault tolerance, Security, Service limits	
Four tag categories	Technical, Automation, Business, Security	

INTEGRATION & ANALYTICS	
Decouple a spiky front end ≈	SQS + auto-scale workers on queue depth
Standard queue guarantees	at-least-once, best-effort order
Exactly-once and strict order	FIFO queue (name ends .fifo)
Message processed twice	visibility timeout shorter than processing
Poison messages / empty receives	dead-letter queue / long polling
One message to many queues	SNS fan-out (+ filter policies)
Filter on event body, SaaS, replay	EventBridge
Coordinate steps, retries, approvals	Step Functions (SWF is legacy)
App already speaks AMQP, MQTT, JMS	Amazon MQ
Many consumers read the same records	Kinesis Data Streams
Load a stream to S3 or Redshift, no code +	Firehose (~60 s, not real time)
Ad-hoc SQL on S3 / cut that bill	Athena / Parquet, compress, partition
Hadoop & Spark / search / BI	EMR / OpenSearch / QuickSight

MONITORING, GOVERNANCE & COST	
CloudWatch / CloudTrail / Config	performance / who did what / configuration and compliance
Metrics needing the agent +	memory and disk space used
Count ERROR lines and alarm	metric filter on a log group
CloudTrail free by default	90 days of management events
Who deleted an S3 object	enable data events
Prove logs untampered / all regions	log file integrity validation / apply the trail everywhere
Fix a non-compliant resource	Config remediation via SSM Automation
Only required CFN section / preview	Resources / change set
Keep data on delete / spot drift	DeletionPolicy Retain / drift detection
Many accounts and regions / guard rails	StackSets / Control Tower + SCPs
Right-size / alert on spend	Compute Optimizer / AWS Budgets
Cut the NAT bill for S3 traffic	S3 gateway endpoint
Survive an AZ vs a region failure	multiple AZs / multiple regions

SCENARIO TRIGGERS		highest marks per minute
"minimal operational overhead"	the serverless or managed option	
"highly available"	multiple AZs	
"disaster recovery", "region outage"	multiple regions	
"cannot be interrupted"	Spot is wrong, however cheap	
"decouple" / "notify many subscribers"	SQS / SNS	
"microseconds" / "sub-millisecond"	DAX / ElastiCache	
"static IP for a firewall allow-list"	NLB or Global Accelerator	
"Windows file share"	FSx for Windows, never EFS	
"no bandwidth, petabytes, deadline"	Snow family	
"urgent" vs "consistent bandwidth"	Site-to-Site VPN / Direct Connect	
"do not open port 22"	Session Manager	
"never hardcode credentials"	an IAM role	
"must be the only key holders"	CloudHSM, or SSE-C for S3	
"longer than 15 minutes"	not Lambda — Fargate, Batch, Step Functions	
"we already use Kubernetes"	EKS	
"must not leave the AWS network"	a VPC endpoint	
"pay per query, nothing running"	Athena	
"petabyte warehouse, high concurrency"	Redshift	
"MongoDB compatible" / "Cassandra"	DocumentDB / Keyspaces	
"no central trusted authority"	Managed Blockchain	
"single pane of glass" for findings	Security Hub	
"socket or core-based licence" +	Dedicated Host	

TIE-BREAKERS, IN ORDER	
1. Managed beats self-managed. 2. Multi-AZ beats single-AZ. 3. Temporary credentials beat long-term keys. 4. A guard rail beats trusting people.	
5. Cheaper wins only if every stated requirement still holds.	
6. Reject the retired: EC2-Classic, Spot block, Classic Load Balancer, SWF.	
7. Reject the impossible: a subnet spanning AZs, an EBS volume across AZs, a CNAME at the zone apex, reading a Multi-AZ standby.	

RESILIENCE & DISASTER RECOVERY	
RTO / RPO	how long you may be down / how much data you may lose
DR strategies, cheapest first	backup and restore, pilot light, warm standby, multi-site active-active
Backup and restore	RTO in hours — only backups in the other region
Pilot light	tens of minutes — data replicating, servers off
Warm standby	minutes — a scaled-down but working copy
Multi-site active-active	near zero — full capacity, both regions serving
Stateless tier survives instance loss	Auto Scaling across AZs behind an ELB
Database survives an AZ loss	Multi-AZ
Database survives a region loss	cross-region replica or Aurora Global Database
Objects survive a region loss	S3 Cross-Region Replication
Static page during an outage	Route 53 failover to an S3 website
Guard against accidental deletion	versioning + MFA Delete + Object Lock

COST OPTIMIZATION		20% of the exam
■ ■ ■ Steady 24/7 workload	Savings Plans or Reserved Instances	
■ ■ ■ Fault-tolerant, interruptible work	Spot Instances	
Idle at night and weekends	scheduled scaling or an instance scheduler	
Over-sized instances ≈	Compute Optimizer	
■ ■ ■ Big NAT bill for S3 traffic	an S3 gateway endpoint	
■ ■ ■ Everything sitting in S3 Standard ≈	lifecycle rules + Intelligent-Tiering	
■ ■ ■ Old snapshots, unattached volumes	find with Trusted Advisor, automate with DLM	
■ ■ ■ Incomplete multipart uploads	lifecycle rule to abort them	
■ ■ ■ High Athena bill	Parquet, compression, partitioning	
■ ■ ■ Spiky DynamoDB traffic	on-demand capacity mode	
Cross-AZ data transfer charges	keep chatty tiers in one AZ	
■ ■ ■ Warn before overspending	AWS Budgets + Cost Anomaly Detection	
■ ■ ■ See cost by team	cost allocation tags + Cost Explorer	

WHAT IS GLOBAL, REGIONAL, OR ZONAL		
■ ■ ■ Global services	IAM, Route 53, CloudFront, Organizations, Billing	
■ ■ ■ Regional	EC2, S3 buckets, RDS, VPC, DynamoDB tables	
■ ■ ■ AZ-scoped	subnets, EC2 instances, EBS volumes, ENIs, NAT gateways	
■ ■ ■ Must be copied cross-region	AMIs, EBS & RDS snapshots, ECR images, KMS keys	
A region has	3 or more AZs, independent power, cooling and network	
Local Zone / Wavelength / Outposts	near a big city / in a 5G network / in your datacentre	
■ ■ ■ Edge locations	CloudFront points of presence — far more numerous than regions	

LOAD BALANCERS IN DEPTH		the single most-asked topic
■ Routes on URL path, host, header, query string	ALB (layer 7)	
■ Routes on TCP/UDP port only	NLB (layer 4)	
■ Preserves the client source IP natively	NLB (ALB uses X-Forwarded-For)	
■ Millions of requests/sec, ultra-low latency	NLB	
■ One static IP per AZ, supports Elastic IPs	NLB	
■ WebSockets, HTTP/2, gRPC	ALB	
■ Redirect HTTP to HTTPS with no app change	ALB listener rule	
■ Return a fixed response, e.g. a maintenance page	ALB listener rule	
■ Authenticate with Cognito or OIDC at the edge	ALB	
■ Route to Lambda as a target	ALB	
■ Route to IP targets, including on-premises	ALB and NLB (IP target type)	
Front a third-party firewall appliance	Gateway Load Balancer (layer 3, GENEVE)	
Health check lives on	the target group, not the listener	
TLS terminates on	the listener, with an ACM certificate	
Several certificates on one listener	SNI	
Sticky sessions	a cookie — duration or application based	
Deregistration delay default	300 s (connection draining)	
■ Spread across AZs but traffic is uneven	turn on cross-zone load balancing (off by default on NLB)	
Trap: an ALB has no static IP — always use its DNS name. If a question needs an address for a customer allow-list, the answer is NLB or Global Accelerator.		

API GATEWAY & FRONT DOORS		4.6% of banked questions
Three API types	REST (full), HTTP (cheaper, faster), WebSocket (two-way)	
Endpoint types	Edge-optimized, Regional, Private	
■ ■ ■ Reachable only inside a VPC	Private endpoint + interface VPC endpoint	
■ ■ ■ Who may call it	IAM auth, Lambda authorizer, Cognito user pool, resource policy	
API key + usage plan is for	throttling and metering per client — not authentication	
Cut repeated backend reads	stage caching	
Protect the backend from spikes	throttling — steady rate plus burst	
Client got a 429	it exceeded the throttle limit	
■ ■ ■ Block SQL injection at the front door	attach a WAF web ACL	
Managed GraphQL with offline sync	AppSync	
■ ■ ■ Sign-up and sign-in for your app users	Cognito user pools	
■ ■ ■ Swap a social login for AWS credentials	Cognito identity pools	

MIGRATION & HYBRID	
Move whole servers to AWS	Application Migration Service (MGN)
■ ■ ■ Move a database with little downtime	DMS (+ SCT to change engine)
Inventory what is running on-premises first	Application Discovery Service
Track a multi-service migration in one place	Migration Hub
Move VMware estate as-is	VMware Cloud on AWS
Keep AWS APIs in your own datacentre	Outposts
Low latency to one metro area	Local Zones
Inside a 5G carrier network	Wavelength
■ ■ ■ Run containers on your own hardware	ECS Anywhere / EKS Anywhere
The 7 Rs	retire, retain, relocate, rehost, replatform, refactor, repurchase

THE LONG TAIL		named in the exam guide, asked rarely
■ ■ ■ Managed Active Directory, or connect to yours	AWS Directory Service (needed by FSx for Windows, WorkSpaces)	
Track BYOL licences against limits	License Manager	
Score an architecture against the pillars	Well-Architected Tool	
Managed Grafana dashboards / Prometheus metrics	Amazon Managed Grafana / Managed Service for Prometheus	
Trace a request across microservices	AWS X-Ray	
■ ■ ■ Ingest and index camera feeds	Kinesis Video Streams	
Move SaaS data to AWS with no code	AppFlow	
Subscribe to third-party datasets	AWS Data Exchange	
Push notifications and campaigns to users	Pinpoint	
Intelligent enterprise search	Kendra	
Approved self-service infrastructure templates	Service Catalog / Proton	
Deliberately break things to test resilience	Fault Injection Simulator	
Virtual desktops / stream one app	WorkSpaces / AppStream 2.0	

ALSO ASKED AS		the same ask, reworded
"minimal operational overhead"	least management effort · most operationally efficient · fully managed · no servers to maintain	
"most cost-effective"	lowest cost · least expensive · minimize spend · MOST cost-efficient	
"highly available"	survive the loss of an AZ · no single point of failure · keep running if one AZ fails	
"disaster recovery"	region-wide outage · RTO and RPO · fail over to another Region · business continuity	
"decouple"	scale the tiers independently · buffer the requests · absorb bursts without losing them	
"lowest latency"	fastest for global users · improve response time · users in Europe report it is slow	
■ ■ ■ "least privilege"	only the permissions required · MOST restrictive · restrict access to only	
"no hardcoded credentials"	without storing access keys · without embedding credentials in the code	
"serverless"	no servers to manage · no infrastructure to maintain · scales automatically	
"cannot be interrupted"	must run to completion · stateful and long-running · cannot tolerate termination	
"must not traverse the internet"	keep traffic on the AWS network · private connectivity · without an internet gateway	
"encrypted at rest"	protected when stored · meet the compliance requirement for stored data	
"improve read performance"	the database is struggling under reads · the same queries repeat · offload reads	
"needs a static IP"	the customer firewall only allows specific IPs · allow-list our address	

MNEMONICS	
Six pillars — Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability.	
Trusted Advisor: CPFSS — Cost, Performance, Fault tolerance, Security, Service limits.	
DR, cheapest first — Backup and restore, Pilot light, Warm standby, Multi-site active-active.	
RTO = Time you may be down. RPO = Point in the past you fall back to.	
Placement groups — Cluster = Close, Spread = Separate, Partition = Pods.	
Default VPC gives five — internet gateway, main route table, security group, network ACL, DHCP option set.	
Groups guard instances. Lists guard subnets. Default NACL is open; a custom NACL is closed.	
Multi-AZ is durability. Read replicas are scalability.	
Geolocation = the user. Geoproximity = your resources.	
Streams if I write the consumer. Firehose if AWS writes it.	
GuardDuty detects. Inspector scans. Macie classifies. Detective investigates. Security Hub collects.	
CloudWatch watches performance. CloudTrail trails API calls. Config tracks configuration.	
Beanstalk deploys the app. CloudFormation declares the infrastructure. OpsWorks is Chef and Puppet.	